



The Effect of Rinsing with Young Coconut Water on Changes in Saliva pH in Elementary School Students

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ABSTRACT

Salivary pH plays an important role in maintaining oral health, particularly in regulating the oral environment and preventing dental caries. Young coconut water contains various minerals and bioactive compounds that may influence oral conditions, including salivary pH. However, evidence regarding the effect of rinsing with young coconut water on salivary pH among elementary school students remains limited. This study aimed to determine the effect of rinsing with young coconut water on changes in salivary pH among sixth-grade students at SD Negeri Kajhu, Baitussalam District, Aceh Besar Regency. A quasi-experimental study with a pretest-posttest non-equivalent control group design was conducted on 60 sixth-grade students, comprising 30 students in the intervention group and 30 students in the control group. The intervention group rinsed with 30 mL of young coconut water for 30 seconds, while the control group rinsed with 30 mL of mineral water for 30 seconds. Salivary pH was measured before and after the procedure using a digital pH meter. Data were analyzed using paired sample t-tests and an independent sample t-test at a significance level of $\alpha = 0.05$. The mean salivary pH in the intervention group decreased from 6.577 before rinsing to 5.940 after rinsing ($p = 0.001$). In the control group, the mean changed from 6.500 to 6.473 ($p = 0.595$). The independent sample t-test showed a significant difference in post-test salivary pH between the groups ($p = 0.001$). Rinsing with young coconut water significantly affected salivary pH and caused a decrease toward a more acidic condition.

INTRODUCTION

Dental caries remains one of the most common oral health problems affecting children worldwide. Elementary school children are particularly vulnerable to dental caries because their oral hygiene practices, dietary habits, and ability to maintain optimal oral health are still developing. Frequent consumption of cariogenic foods and beverages, inadequate toothbrushing practices, and prolonged exposure of teeth to an acidic oral environment may increase the risk of dental caries. Therefore, maintaining a favorable oral environment is an important component of dental caries prevention among school-aged children. Saliva plays an essential role in maintaining oral health. It contributes to oral homeostasis through several mechanisms, including lubrication, buffering of acids, remineralization of dental enamel, and clearance of food debris and microorganisms. One of the important indicators of the oral environment is salivary pH. Under normal conditions, saliva helps neutralize acids produced by oral microorganisms and acids originating from dietary intake (Mokoginta & Wowor, 2017; Arinawati & Widyawati, 2022).

A decrease in salivary pH can create a more favorable environment for enamel demineralization and may contribute to the development and progression of dental caries. Consequently, interventions that can help maintain or restore salivary pH may have potential value in promoting oral health among children. A lower salivary pH indicates a more acidic oral environment and may be associated with caries risk (Zhafirah et al., 2020; Kim et al., 2021).

Rinsing the mouth with certain natural substances has been explored as a simple approach to supporting oral hygiene and maintaining oral environmental conditions. One natural substance that has attracted attention is young coconut water. Young coconut water is a natural beverage containing water, electrolytes, minerals, carbohydrates, and various bioactive components. Its mineral content, particularly potassium and other electrolytes, has been associated with several physiological functions. From an oral health perspective, the use of young coconut water as a mouth-rinsing agent is of interest because it may influence the physicochemical characteristics of saliva and potentially contribute to maintaining salivary

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pH. Young coconut water contains water, carbohydrates, vitamins, minerals, amino acids, and organic components, and its relatively acidic pH may influence salivary pH after oral exposure (Ibrahim, 2020; Mokoginta & Wowor, 2017).

Previous studies have reported that gargling with young coconut water can change salivary pH. Mokoginta and Wowor (2017) reported a significant decrease in salivary pH after gargling with young coconut water. Similar findings were reported by Burhan (2020) and Yuni and Hamid (2024). Based on this background, this study aimed to determine the effect of rinsing with young coconut water on changes in salivary pH among sixth-grade students at SD Negeri Kajhu, Baitussalam District, Aceh Besar Regency

METHOD

This study used a quasi-experimental design with a pretest-posttest non-equivalent control group design. The study was conducted at SD Negeri Kajhu, Baitussalam District, Aceh Besar Regency, on 9 May 2026. The population consisted of 60 sixth-grade students, and 60 students participated as the sample. The thesis described random sampling with inclusion criteria including willingness to participate and ability to cooperate during the procedure.

Data were collected directly from participants. Parents provided informed consent one day before the study. Salivary pH was measured using a digital pH meter. Before measurement, participants were instructed to collect saliva at the floor of the mouth for 60 seconds and then expectorate it into a prepared container. The intervention group rinsed with 30 mL of young coconut water for 30 seconds, while the control group rinsed with 30 mL of Aqua mineral water for 30 seconds. Saliva was collected again for 60 seconds after rinsing and the pH was measured. The 30-second rinsing procedure was retained from the thesis protocol.

Data analysis included descriptive frequency distributions, Shapiro-Wilk normality testing, paired sample t-tests for within-group comparisons, and an independent sample t-test for the post-test comparison between groups. Statistical significance was set at $\alpha = 0.05$

RESULT

Table 1. Frequency Distribution on Characteristic Respondent

Characteristic	Intervention n (%)	Control n (%)	Total n	Total (%)
Age 11 years	6 (20.0)	7 (23.3)	13	21.7
Age 12 years	24 (80.0)	23 (76.7)	47	78.3
Male	13 (43.3)	16 (53.3)	29	48.3
Female	17 (56.7)	14 (46.7)	31	51.7

The table shows the frequency distribution of respondents according to age and sex in the intervention and control groups. Of the 60 respondents, 13 (21.7%) were 11 years old and 47 (78.3%) were 12 years old. In the intervention group, 6 respondents (20.0%) were 11 years old and 24 (80.0%) were 12 years old, while in the control group, 7 respondents (23.3%) were 11 years old and 23 (76.7%) were 12 years old. Regarding sex, the respondents consisted of 29 males (48.3%) and 31 females (51.7%). The intervention group comprised 13 males (43.3%) and 17 females (56.7%), whereas the control group comprised 16 males (53.3%) and 14 females (46.7%). Overall, the distribution of age and sex between the intervention and control groups was relatively comparable, with the majority of respondents being 12 years old and female.

Table 2. Salivary pH before and after rinsing

Group	Pre-test mean $\pm$ SD	Post-test mean $\pm$ SD	Mean change	p-value
Young coconut water	6.577 $\pm$ 0.3794	5.940 $\pm$ 0.2458	0.6367	0.001
Mineral water	6.500 $\pm$ 0.2586	6.473 $\pm$ 0.1893	0.0267	0.595
Post-test between groups	—	5.940 vs 6.473	0.5333	0.001

Based on Table 2, there was the intervention group, 19 students (63.3%) had acidic salivary pH and 11 students (36.7%) had neutral pH before rinsing. After rinsing, all 30 students (100%) were categorized as having acidic salivary pH. In the control group, 23 students (76.7%) had acidic pH and 7 students (23.3%) had neutral pH before rinsing; after rinsing, 28 students (93.3%) had acidic pH and 2 students (6.7%) had neutral pH. The intervention group's mean salivary pH decreased from 6.577 to 5.940, with a mean difference of 0.6367 and $p = 0.001$. The control group's mean changed from 6.500 to 6.473, with a mean difference of 0.0267 and $p = 0.595$. The independent sample t-test showed a significant difference in post-test pH between the intervention and control groups (5.940 vs 6.473; mean difference 0.5333; $p = 0.001$).

DISCUSSION

The findings showed that rinsing with 30 mL of young coconut water for 30 seconds significantly decreased salivary pH. The mean pH fell from 6.577 to 5.940 and the within-group difference was statistically significant ($p = 0.001$). This result was consistent with earlier studies reporting a decrease in salivary pH after exposure to young coconut water, including Mokoginta and Wowor (2017), Burhan (2020), and Yuni and Hamid (2024). A recent study by Abdi et al. (2025) also reported a significant reduction in salivary pH after gargling with green coconut water.

One possible explanation is the acidic nature of the young coconut water used in this study. The thesis reported that the measured pH of the young coconut water used by the researcher was 5.0. Young coconut water may contain organic acids, vitamin C, reducing sugars, and other components that can influence salivary pH. Therefore, the immediate fall in pH observed after rinsing may reflect direct contact between the acidic liquid and saliva rather than an antibacterial effect.

Although young coconut water contains tannins that may have antibacterial potential, an antibacterial effect does not necessarily mean that salivary pH will increase. In this study, pH was measured immediately after rinsing, so the direct effect of the acidic liquid was likely more prominent than any longer-term biological effect. The finding therefore supports the observed decrease in salivary pH under the conditions tested.

Saliva has a buffering system that contributes to maintaining oral pH. Because pH was measured immediately after rinsing, the measurement captured the early response after exposure. A longer interval could produce a different value because salivary buffering may gradually restore pH. Salivary flow and buffering capacity were not directly measured in this study, so their contribution to the observed change could not be determined.

In the control group, rinsing with mineral water produced only a small, non-significant change in mean pH from 6.500 to 6.473 ($p = 0.595$). The significant post-test difference between groups further indicates that the change in the intervention group was greater than the change observed after rinsing with mineral water. The results should nevertheless be interpreted within the study conditions and should not be generalized to longer-term oral-health outcomes

CONCLUSION

Rinsing with young coconut water significantly affected salivary pH in sixth-grade students at SD Negeri Kajhu. The mean salivary pH decreased from 6.577 before rinsing to 5.940 after rinsing ($p = 0.001$), while the control group showed no significant change (6.500 to 6.473; $p = 0.595$). The post-test difference between groups was also significant ($p = 0.001$), indicating that under the conditions of this study, young coconut water caused salivary pH to become more acidic. Further studies may consider larger samples, different observation intervals, and factors such as salivary flow and buffering capacity.

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